

Gene expression modifications

Transgenic strategies in mice involve:

- Random insertion of gene into mouse genome
- Targeted insertion or deletion of gene through homologous recombination with mouse gene

Knock-**out** = removing a gene, taking it **out**.

Knock-**in** = **in**serting a gene.

Random insertion—constitutive expression.

Targeted insertion—conditional expression.

RNA interference

Process whereby small non-coding RNA molecules target mRNAs to inhibit gene expression.

MicroRNA

Naturally produced by cell as hairpin structures. Loose nucleotide pairing allows broad targeting of related mRNAs. When miRNA binds to mRNA, it blocks translation of mRNA and sometimes facilitates its degradation.

Abnormal expression of miRNAs contributes to certain malignancies (eg, by silencing an mRNA from a tumor suppressor gene).

Small interfering RNA

Usually derived from exogenous dsRNA source (eg, virus). Once inside a cell, siRNA requires complete nucleotide pairing, leading to highly specific mRNA targeting. Results in mRNA cleavage prior to translation.

Can be produced by transcription or chemically synthesized for gene “knockdown” experiments.

► BIOCHEMISTRY—GENETICS**Genetic terms**

TERM	DEFINITION	EXAMPLE
Codominance	Both alleles contribute to the phenotype of the heterozygote.	Blood groups A, B, AB; α_1 -antitrypsin deficiency; HLA groups.
Variable expressivity	Patients with the same genotype have varying phenotypes.	Two patients with neurofibromatosis type 1 (NF1) may have varying disease severity.
Incomplete penetrance	Not all individuals with a pathogenic gene variant show the disease. % penetrance $\times$ probability of inheriting genotype = risk of expressing phenotype.	BRCA1 gene mutations do not always result in breast or ovarian cancer.
Pleiotropy	One gene contributes to multiple phenotypic effects.	Cystic fibrosis manifests with thick mucus in the lungs and GI tract, pancreatic insufficiency, male infertility.
Anticipation	Increased severity or earlier onset of disease in succeeding generations.	Trinucleotide repeat diseases (eg, Huntington disease).
Loss of heterozygosity	If a patient inherits or develops a mutation in a tumor suppressor gene, the wild type allele must be deleted/mutated/eliminated before cancer develops. This is not true of oncogenes.	Retinoblastoma and the “two-hit hypothesis,” Lynch syndrome (HNPCC), Li-Fraumeni syndrome.
Epistasis	The allele of one gene affects the phenotypic expression of alleles in another gene.	Albinism, alopecia.
Aneuploidy	An abnormal number of chromosomes; due to chromosomal nondisjunction during mitosis or meiosis.	Down syndrome (trisomy 21), Turner syndrome (45,XO), oncogenesis.

Genetic terms (continued)

TERM	DEFINITION	EXAMPLE
Dominant negative mutation	Exerts a dominant effect. A heterozygote produces a nonfunctional altered protein that also prevents the normal gene product from functioning.	A single mutated <i>p53</i> tumor suppressor gene results in a protein that is able to bind DNA and block the wild type <i>p53</i> from binding to the promoter.
Linkage disequilibrium	Tendency for certain alleles to occur in close proximity on the same chromosome more or less often than expected by chance. Measured in a population, not in a family, and often varies in different populations.	<i>HLA</i> gene, <i>CFTR</i> gene.
Mosaicism	Presence of genetically distinct cell lines in the same individual. Somatic mosaicism—mutation arises from mitotic errors after fertilization and propagates through multiple tissues or organs. Germline (gonadal) mosaicism—mutation only in egg or sperm cells. If parents and relatives do not have the disease, suspect gonadal (or germline) mosaicism.	McCune-Albright syndrome —due to G_s -protein (<i>GNAS</i>) activating mutation. Presents with unilateral café-au-lait spots A with ragged edges, polyostotic fibrous dysplasia (bone is replaced by collagen and fibroblasts), and at least one endocrinopathy (eg, precocious puberty). Lethal if mutation occurs before fertilization (affecting all cells), but survivable in patients with mosaicism.
Locus heterogeneity	Mutations at different loci result in the same disease.	Albinism, retinitis pigmentosa, familial hypercholesteremia.
Allelic heterogeneity	Different mutations in the same locus result in the same disease.	β -thalassemia.
Heteroplasmy	Presence of both normal and mutated mtDNA, resulting in variable expression in mitochondrially inherited disease.	mtDNA passed from mother to all children. Example: Leber hereditary optic neuropathy.
Uniparental disomy	Offspring receives 2 copies of a chromosome from 1 parent and no copies from the other parent. HeterodIsomy (heterozygous) indicates a meiosis I error. IsodIsomy (homozygous) indicates a meiosis II error or postzygotic chromosomal duplication of one of a pair of chromosomes, and loss of the other of the original pair.	Uniparental is euploid (correct number of chromosomes). Most occurrences of uniparental disomy (UPD) → normal phenotype. Consider isodisomy in an individual manifesting a recessive disorder when only one parent is a carrier. Examples: Prader-Willi and Angelman syndromes.

Population genetics

CONCEPT	DESCRIPTION	EXAMPLE
Bottleneck effect	Fitness equal across alleles → natural disaster that removes certain alleles by chance → new allelic frequency (by chance, not naturally selected).	The founder effect is a type of bottleneck when cause is due to calamitous population separation.
Natural selection	Alleles that increase species fitness are more likely to be passed down to offspring and vice versa.	Human evolution.
Genetic drift	Also called allelic drift or Wright effect. A dramatic shift in allelic frequency that occurs by chance (not by natural selection).	Founder effect and bottleneck effect are both examples of genetic drift.

Hardy-Weinberg principle

	A (p)	a (q)
A (p)	AA (p ²)	Aa (pq)
a (q)	Aa (pq)	aa (q ²)

In a given population where mating is at random, allele and genotype frequencies will be constant. If **p** and **q** represent the frequencies of alleles A and a in a population, respectively, then **p + q = 1**, where:

- **p²** = frequency of homozygosity for allele A
- **q²** = frequency of homozygosity for allele a
- **2pq** = frequency of heterozygosity (carrier frequency, if an autosomal recessive disease)

Therefore the sum of the frequencies of these genotypes is **p² + 2pq + q² = 1**.

The frequency of an X-linked recessive disease in males = **q** and in females = **q²**.

Hardy-Weinberg law assumptions include:

- No mutation occurring at the locus
- Natural selection is not occurring
- Completely random mating
- No net migration
- Large population

If a population is in Hardy-Weinberg equilibrium, then the values of **p** and **q** remain constant from generation to generation.

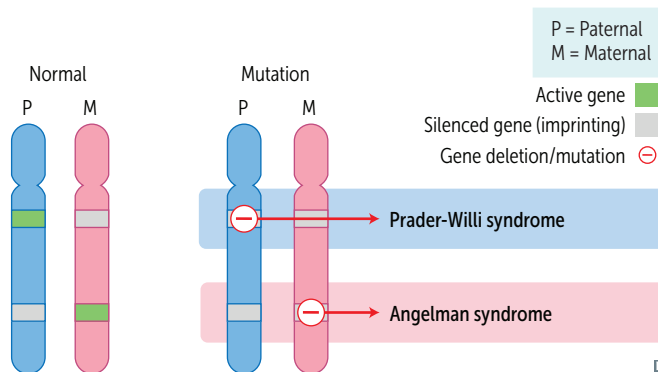
In rare autosomal recessive diseases, **p ≈ 1**.

Example: The prevalence of cystic fibrosis (an autosomal recessive disease) in the US is approximately 1/3200, which tells us that **q² = 1/3200**, with **q ≈ 0.017** or 1.7% of the population. Since **p + q = 1**, we know that **p = 1 - √(1/3200) ≈ 0.982**, which gives us a heterozygous carrier frequency of **2pq = 0.035** or 3.5% of the population. Notice that since the disease is relatively rare, we could have approximated **p ≈ 1** and obtained a similar result.

Disorders of imprinting

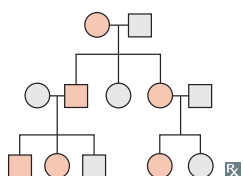
One gene copy is silenced by methylation, and only the other copy is expressed → parent-of-origin effects. The expressed copy may be mutated, may not be expressed, or may be deleted altogether.

	Prader-Willi syndrome	Angelman syndrome
WHICH GENE IS SILENT?	Maternally derived genes are silenced (except UBE3A) Disease occurs when the p aternal allele is deleted or mutated	Paternally derived UBE3A is silenced Disease occurs when the m aternal allele is deleted or mutated
SIGNS AND SYMPTOMS	Hyperphagia, obesity, intellectual disability, hypogonadism, hypotonia	H and-flapping, A taxia, severe I ntellectual disability, inappropriate L aughter, S eizures. HAILS the Angels.
CHROMOSOMES INVOLVED	Chromosome 15 of paternal origin	UBE3A on maternal copy of chromosome 15
NOTES	25% of cases are due to maternal uniparental disomy POP : P radler-Willi, O besity/overeating, P aternal allele deleted	5% of cases are due to paternal uniparental disomy MAMAS : M aternal allele deleted, A ngelman syndrome, M ood, A taxia, S eizures



Modes of inheritance

Autosomal dominant

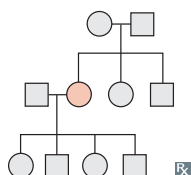


Often due to defects in structural genes. Many generations, both males and females are affected.

	A	a
a	Aa	aa
a	Aa	aa

Often pleiotropic (multiple apparently unrelated effects) and variably expressive (different between individuals). Family history crucial to diagnosis. With one affected (heterozygous) parent, each child has a 50% chance of being affected.

Autosomal recessive



With 2 carrier (heterozygous) parents, on average: each child has a 25% chance of being affected, 50% chance of being a carrier, and 25% chance of not being affected nor a carrier.

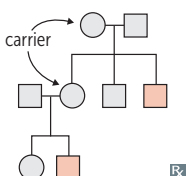
	A	a
A	AA	Aa
a	Aa	aa

Often due to enzyme deficiencies. Usually seen in only 1 generation. Commonly more severe than dominant disorders; patients often present in childhood.

↑ risk in consanguineous families.

Unaffected individual with affected sibling has 2/3 probability of being a carrier.

X-linked recessive

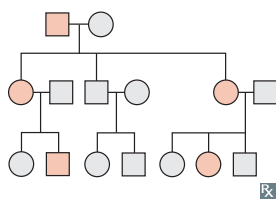


Sons of heterozygous mothers have a 50% chance of being affected. No male-to-male transmission. Skips generations.

	X	X	X	X
X	XX	XX	X	XX
Y	XY	XY	Y	XY

Commonly more severe in males. Females usually must be homozygous to be affected.

X-linked dominant

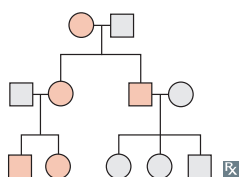


Transmitted through both parents. Children of affected mothers each have a 50% chance of being affected. 100% of daughters and 0% of sons of affected fathers will be affected.

	X	X	X	X
X	XX	XX	X	XX
Y	XY	XY	Y	XY

Examples: fragile X syndrome, Alport syndrome, **hypophosphatemic rickets** (also called X-linked hypophosphatemia)—phosphate wasting at proximal tubule → ricketslike presentation.

Mitochondrial inheritance



Transmitted only through the mother. All offspring of affected females may show signs of disease.

Variable expression in a population or even within a family due to heteroplasmy.

Caused by mutations in mtDNA.

Examples: mitochondrial myopathies, Leber hereditary optic neuropathy.

□ = unaffected male; ■ = affected male; ○ = unaffected female; ● = affected female.

Autosomal dominant diseases

Achondroplasia, autosomal dominant polycystic kidney disease, familial adenomatous polyposis, familial hypercholesterolemia, hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu syndrome), hereditary spherocytosis, Huntington disease, Li-Fraumeni syndrome, Marfan syndrome, multiple endocrine neoplasias, myotonic muscular dystrophy, neurofibromatosis type 1 (von Recklinghausen disease), neurofibromatosis type 2, tuberous sclerosis, von Hippel-Lindau disease.

Autosomal recessive diseases

Mostly consist of enzyme defects. Oculocutaneous albinism, phenylketonuria, cystic fibrosis, sickle cell disease, Wilson disease, sphingolipidoses (except Fabry disease), hemochromatosis, glycogen storage diseases, thalassemia, mucopolysaccharidoses (except Hunter syndrome), Friedrich ataxia, Kartagener syndrome, ARPKD. Oh, please! Can students who score high grades tell me features of the kidney disorder **Autosomal Recessive Polycystic Kidney Disease**?

Cystic fibrosis**GENETICS**

Autosomal recessive; defect in CFTR gene on chromosome 7 (deletion; $\Delta F508$). Most common lethal genetic disease in patients with European ancestry.

PATHOPHYSIOLOGY

CFTR encodes an ATP-gated Cl^- channel (secretes Cl^- in lungs/GI tract, reabsorbs Cl^- in sweat glands). Phe508 deletion $\rightarrow$ misfolded protein $\rightarrow$ improper protein trafficking $\rightarrow$ protein absent from cell membrane $\rightarrow$ $\downarrow$ Cl^- (and H_2O and HCO_3^-) secretion $\rightarrow$ compensatory $\uparrow$ Na^+ reabsorption via epithelial Na^+ channels (ENaC) $\rightarrow$ $\uparrow$ H_2O reabsorption $\rightarrow$ abnormally thick mucus secreted into lungs/GI tract. $\uparrow$ Na^+ reabsorption $\rightarrow$ more negative transepithelial potential difference.

DIAGNOSIS

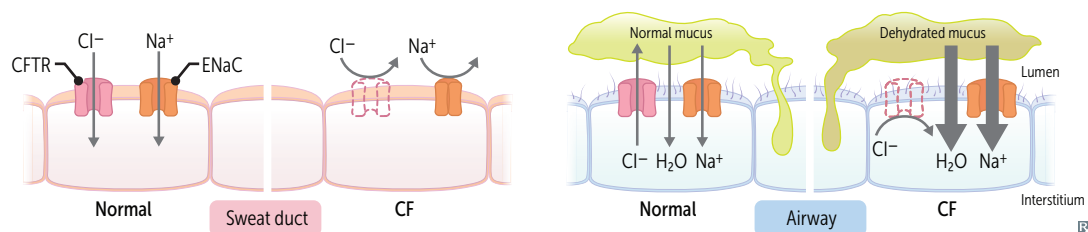
$\uparrow$ Cl^- concentration in pilocarpine-induced sweat test. Can present with contraction alkalosis and hypokalemia (ECF effects analogous to loop diuretic effect) due to ECF $\text{H}_2\text{O}/\text{Na}^+$ losses via sweating and concomitant renal K^+/H^+ wasting. $\uparrow$ immunoreactive trypsinogen (newborn screening) due to clogging of pancreatic duct.

COMPLICATIONS

Recurrent pulmonary infections (eg, *S aureus* [infancy and early childhood], *P aeruginosa* [adulthood], allergic bronchopulmonary aspergillosis [ABPA]), chronic bronchitis and bronchiectasis $\rightarrow$ reticulonodular pattern on CXR, opacification of sinuses. Nasal polyps, nail clubbing. Pancreatic insufficiency, malabsorption with steatorrhea, and fat-soluble vitamin deficiencies (A, D, E, K) progressing to endocrine dysfunction (CF-related diabetes), biliary cirrhosis, liver disease. Meconium ileus in newborns. Infertility in males (absence of vas deferens, spermatogenesis may be unaffected) and subfertility in females (amenorrhea, abnormally thick cervical mucus).

TREATMENT

Multifactorial: chest physiotherapy, aerosolized dornase alfa (DNase), and inhaled hypertonic saline $\rightarrow$ mucus clearance. Azithromycin prevents acute exacerbations. Ibuprofen for anti-inflammatory effect. Pancreatic enzyme replacement therapy (pancrelipase) for pancreatic insufficiency. CFTR modulators can be used alone or in combination. Efficacy varies by different genetic mutations (pharmacogenomics). Are either potentiators (hold gate of CFTR channel open $\rightarrow$ Cl^- flows through cell membrane; eg, ivacaftor) or correctors (help CFTR protein to form right 3-D shape $\rightarrow$ moves to the cell surface; eg, lumaCaFToR, tezaCaFToR).



X-linked recessive diseases

Bruton agammaglobulinemia, Duchenne and Becker muscular dystrophies, Fabry disease, G6PD deficiency, hemophilia A and B, Hunter syndrome, Lesch-Nyhan syndrome, ocular albinism, ornithine transcarbamylase (OTC) deficiency, Wiskott-Aldrich syndrome.

Females with Turner syndrome (45,XO) are more likely to have an X-linked recessive disorder.

X-inactivation (lyonization)—during development, one of the X chromosomes in each XX cell is randomly deactivated and condensed into a Barr body (methylated heterochromatin). If skewed inactivation occurs, XX individuals may express X-linked recessive diseases (eg, G6PD); penetrance and severity of X-linked dominant diseases in XX individuals may also be impacted.

Muscular dystrophies**Duchenne**

X-linked recessive disorder typically due to **frameshift** deletions or nonsense mutations → truncated or absent dystrophin protein → progressive myofiber damage. Can also result from splicing errors.

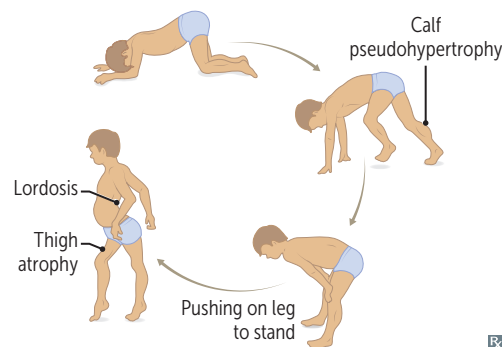
Weakness begins in pelvic girdle muscles and progresses superiorly. Pseudohypertrophy of calf muscles due to fibrofatty replacement of muscle **A**. Waddling gait.

Onset before 5 years of age. **D**ilated cardiomyopathy is common cause of **d**death.

Gowers sign—patient uses upper extremities to help stand up. Classically seen in Duchenne muscular dystrophy, but also seen in other muscular dystrophies and inflammatory myopathies (eg, polymyositis).

Duchenne = **d**eleted **d**ystrophin.

Dystrophin gene (*DMD*) is the largest protein-coding human gene → ↑ chance of spontaneous mutation. Dystrophin helps to anchor muscle fibers to the extracellular matrix, primarily in skeletal and cardiac muscles. Loss of dystrophin → myonecrosis. ↑ CK and aldolase; genetic testing confirms diagnosis.

**Becker**

X-linked recessive disorder typically due to **non-frameshift** deletions in dystrophin gene (partially functional instead of truncated). Less severe than Duchenne (**B**ecker is **b**etter). Onset in adolescence or early adulthood.

Deletions can cause both Duchenne and Becker muscular dystrophies. 2/3 of cases have large deletions spanning one or more exons.

Myotonic dystrophy

Autosomal dominant. Onset 20–30 years. **CTG** trinucleotide repeat expansion in the *DMPK* gene → abnormal expression of myotonin protein kinase → percussion myotonia (eg, difficulty releasing hand from handshake), muscle wasting, cataracts, testicular atrophy, frontal balding, arrhythmia.

Cataracts, **T**oupee (early balding in males), **G**onadal atrophy. Muscle biopsy shows ring fibers and central nuclei.

Mitochondrial diseases Rare disorders arising 2° to failure in oxidative phosphorylation. Tissues with ↑ energy requirements are preferentially affected (eg, CNS, skeletal muscle).

Mitochondrial myopathies—include **MELAS** (mitochondrial **e**ncephalomyopathy with **l**actic **a**cidosis and **s**tokelike episodes) and **MERRF** (**m**yoclonic **e**pilepsy with **r**agged **r**ed **f**ibers). Light microscopy with stain: ragged red fibers due to compensatory proliferation of mitochondria. Electron microscopy: mitochondrial crystalline inclusions.

Leber hereditary optic neuropathy—mutations in complex I of ETC → neuronal death in retina and optic nerve → subacute bilateral vision loss in teens/young adults (males > females). Usually permanent. May be accompanied by neurologic dysfunction (eg, tremors, multiple sclerosis–like illness).

Rett syndrome Sporadic disorder caused by de novo mutation of *MECP2* on X chromosome. Seen mostly in females. Embryonically lethal in males. Individuals with **Rett** syndrome experience initial normal development (6–18 months) followed by regression (“**re**ttur”) in motor, verbal, and cognitive abilities; ataxia; seizures; scoliosis; and stereotypic hand-wringing.

Fragile X syndrome X-linked (atypical) inheritance. Trinucleotide repeats in *FMR1* → hypermethylation of cytosine residues → ↓ expression. Most common inherited cause of intellectual disability (Down syndrome is most common genetic cause, but most cases occur sporadically).

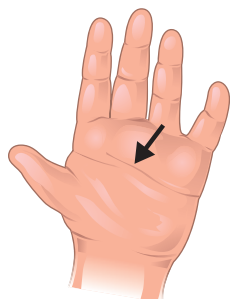
Trinucleotide repeat expansion [(CGG)_n] occurs during oogenesis.
 Premutation (50–200 repeats) → tremor, ataxia, 1° ovarian insufficiency.
 Full mutation (>200 repeats) → postpubertal macroorchidism (enlarged testes), long face with large jaw, large everted ears, autism, mitral valve prolapse, hypermobile joints. Self-mutilation is common and can be confused with Lesch-Nyhan syndrome.

Trinucleotide repeat expansion diseases May show genetic anticipation (disease severity ↑ and age of onset ↓ in successive generations).

DISEASE	TRINUCLEOTIDE REPEAT	MODE OF INHERITANCE	MNEMONIC
Huntington disease	(CAG) _n	AD	C audate has ↓ A Ch and G A _B A
Myotonic dystrophy	(CTG) _n	AD	C ataracts, T oupee (early balding in males), G onadal atrophy in males, reduced fertility in females
Fragile X syndrome	(CGG) _n	XD	C hin (protruding), G iant G onads
Friedreich ataxia	(GAA) _n	AR	Ataxic GAA it

Autosomal trisomies

Autosomal trisomies are screened in first and second trimesters with noninvasive prenatal tests. Incidence of trisomies: Down (21) > Edwards (18) > Patau (13). Autosomal monosomies are incompatible with life (high chance of recessive trait expression).

Down syndrome (trisomy 21)

Single palmar crease

Findings: intellectual disability, flat facies, prominent epicanthal folds, single palmar crease, incurved 5th finger, gap between 1st 2 toes, duodenal atresia, Hirschsprung disease, congenital heart disease (eg, AVSD), Brushfield spots (whitish spots at the periphery of the iris). Associated with early-onset Alzheimer disease (chromosome 21 codes for amyloid precursor protein), ↑ risk of AML/ALL.

95% of cases due to meiotic nondisjunction, most commonly during meiosis I (↑ with advanced maternal age: from 1:1500 in females < 20 to 1:25 in females > 45). 4% of cases due to unbalanced Robertsonian translocation, most typically between chromosomes 14 and 21. 1% of cases due to postfertilization mitotic error.

Drinking age (21).

Most common viable chromosomal disorder and most common cause of genetic intellectual disability.

First-trimester ultrasound commonly shows ↑ nuchal translucency and hypoplastic nasal bone. Markers are **hi** up: ↑ **h**CG, ↑ **i**nhibin. ↑ risk of umbilical hernia (incomplete closure of umbilical ring).

The **6 As** of Down syndrome:

- **A**tlantoaxial instability
- **A**dvanced maternal age
- **A**tresia (duodenal)
- **A**trioventricular septal defect
- **A**lzheimer disease (early onset)
- **A**ML (<5 years of age)/**A**LL (>5 years of age)

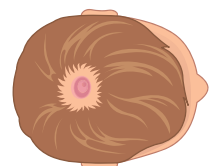
Edwards syndrome (trisomy 18)

Findings: **PRINCE** Edward—**P**rominent occiput, **R**ocker-bottom feet, **I**ntellectual disability, **N**ondisjunction, **C**lenched fists with overlapping fingers, low-set **E**ars, micrognathia (small jaw), congenital heart disease (eg, VSD), omphalocele, myelomeningocele. Death usually occurs by age 1.

Election age (18).

2nd most common autosomal trisomy resulting in live birth (most common is Down syndrome).

In **E**dwards syndrome, **e**very prenatal screening marker **d**ecreases.

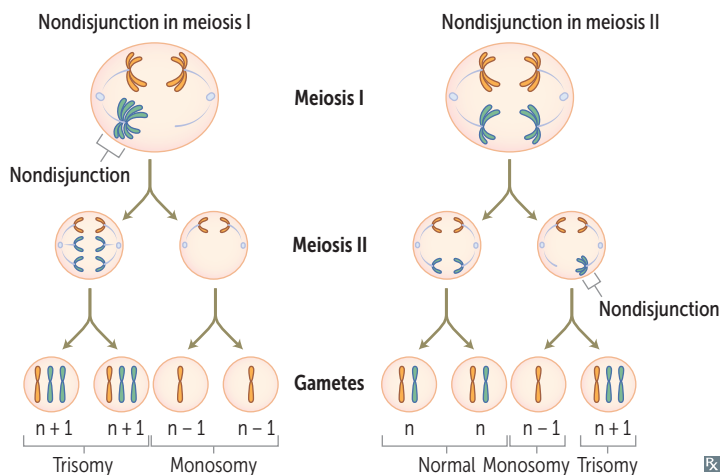
Patau syndrome (trisomy 13)

Cutis aplasia

Findings: severe intellectual disability, rocker-bottom feet, microphthalmia, microcephaly, cleft lip/palate, holoprosencephaly, polydactyly, cutis aplasia, congenital heart (pump) disease, polycystic kidney disease, omphalocele. Death usually occurs by age 1.

Puberty at age 13.

Defect in fusion of prechordal mesoderm → midline defects.



1st trimester screening		
Trisomy	β -hCG	PAPP-A
21	↑	↓
18	↓	↓
13	↓	↓

2nd trimester (quadruple) screening				
Trisomy	hCG	Inhibin A	Estriol	AFP
21	↑	↑	↓	↓
18	↓	— or ↓	↓	↓
13	—	—	—	—

Genetic disorders by chromosome

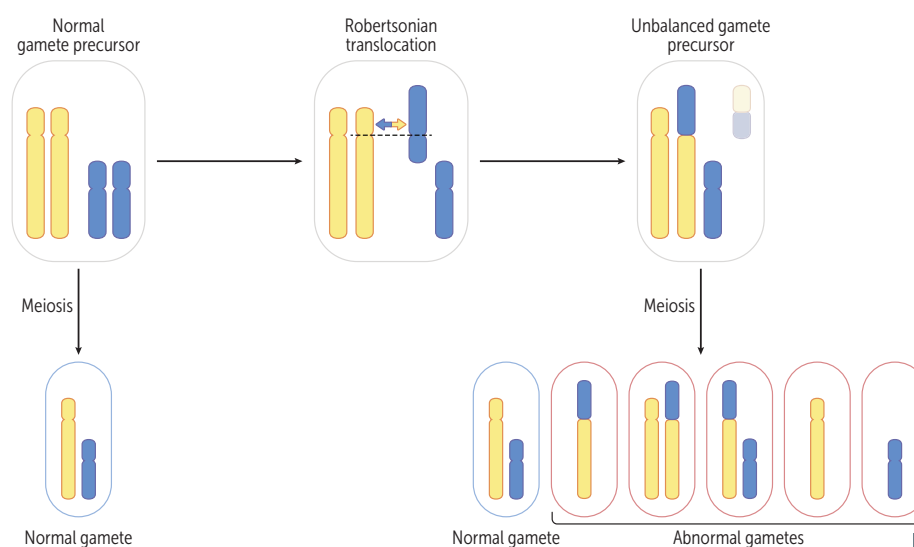
CHROMOSOME	SELECTED EXAMPLES
3	von Hippel-Lindau disease, renal cell carcinoma
4	ADPKD (<i>PKD2</i>), achondroplasia, Huntington disease
5	Cri-du-chat syndrome, familial adenomatous polyposis
6	Hemochromatosis (<i>HFE</i>)
7	Williams syndrome, cystic fibrosis
9	Friedreich ataxia, tuberous sclerosis (<i>TSC1</i>)
11	Wilms tumor, β -globin gene defects (eg, sickle cell disease, β -thalassemia), MEN1
13	Patau syndrome, Wilson disease, retinoblastoma (<i>RBI</i>), <i>BRCA2</i>
15	Prader-Willi syndrome, Angelman syndrome, Marfan syndrome
16	ADPKD (<i>PKD1</i>), α -globin gene defects (eg, α -thalassemia), tuberous sclerosis (<i>TSC2</i>)
17	Neurofibromatosis type 1, <i>BRCA1</i> , <i>TP53</i> (Li-Fraumeni syndrome)
18	Edwards syndrome
21	Down syndrome
22	Neurofibromatosis type 2, DiGeorge syndrome (22q11)
X	Fragile X syndrome, Turner syndrome (XO), XLA, Klinefelter syndrome (XXY)

Robertsonian translocation

Chromosomal translocation that commonly involves chromosome pairs **21**, **22**, **13**, **14**, and **15**.

One of the most common types of translocation. Occurs when the long arms of 2 acrocentric chromosomes (chromosomes with centromeres near their ends) fuse at the centromere and the 2 short arms are lost.

Balanced translocations (no gain or loss of significant genetic material) normally do not cause abnormal phenotype. Unbalanced translocations (missing or extra genes) can result in miscarriage, stillbirth, and chromosomal imbalance (eg, Down syndrome, Patau syndrome).

**Cri-du-chat syndrome**

Cri du chat = cry of the cat. Congenital deletion on short arm of chromosome 5 (46,XX or XY, 5p-). Findings: microcephaly, moderate to severe intellectual disability, high-pitched **c**rying, epicanthal folds, cardiac abnormalities (**VSD**). I **c**ry when I am **V**ery **SaD**.

Williams syndrome

Congenital microdeletion of long arm of chromosome 7 (deleted region includes elastin gene). Findings: distinctive “elfin” facies, intellectual disability, hypercalcemia, well-developed verbal skills, extreme friendliness with strangers, cardiovascular problems (eg, supraventricular aortic stenosis, pulmonary artery stenosis, renal artery stenosis).

► BIOCHEMISTRY—NUTRITION

Essential fatty acids

Polyunsaturated fatty acids that cannot be synthesized in the body and must be provided in the diet (eg, nuts/seeds, plant oils, seafood). Linoleic acid (omega-6) is metabolized to arachidonic acid, which serves as the precursor to leukotrienes and prostaglandins. Linolenic acid (omega-3) and its metabolites have cardioprotective and antihyperlipidemic effects.

In contrast, consumption of *trans*-unsaturated fatty acids (found in fast food) promotes cardiovascular disease by ↑ LDL and ↓ HDL.

Vitamins: fat soluble

A, D, E, K. Absorption dependent on bile emulsification, pancreatic secretions, and intact ileum. Toxicity more common than for water-soluble vitamins because fat-soluble vitamins accumulate in fat.

Malabsorption syndromes with steatorrhea (eg, cystic fibrosis and celiac disease) or mineral oil intake can cause fat-soluble vitamin deficiencies.

Vitamins: water soluble

B₁ (thiamine: TPP)
B₂ (riboflavin: FAD, FMN)
B₃ (niacin: NAD⁺)
B₅ (pantothenic acid: CoA)
B₆ (pyridoxine: PLP)
B₇ (biotin)
B₉ (folate)
B₁₂ (cobalamin)
C (ascorbic acid)

Wash out easily from body except B₁₂ and B₉. B₁₂ stored in liver for ~ 3–4 years. B₉ stored in liver for ~ 3–4 months. B-complex deficiencies often result in dermatitis, glossitis, and diarrhea. Can be coenzymes (eg, ascorbic acid) or precursors to coenzymes (eg, FAD, NAD⁺).

Dietary supplementation

DIET	SUPPLEMENTATION REQUIRED
Vegetarian/vegan	Vitamin B ₁₂ Iron Vitamin B ₂ Frequently, vitamin D (although this is commonly deficient in many diets)
High egg white (raw)	Vitamin B ₇ (avidin in egg whites binds biotin and prevents absorption)
Untreated corn	Vitamin B ₃ (deficiency is common in resource-limited areas)